

Lucas Ang

NUS Merit Scholar | Business Analytics with Economics | Applied AI, Fintech ML, Full-Stack Systems

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Education

National University of Singapore - BSc (Business Analytics with Economics)

2024 - 2027

- Awarded NUS Merit Scholarship for academic excellence and leadership potential.

Raffles Institution - Integrated Programme, Singapore-Cambridge GCE 'A' Levels

2016 - 2021

- Awarded Edusave Scholarship for Independent Schools for academic excellence.

Experience

Central Provident Fund Board - Applied AI Engineer Intern

Jun 2026 - Dec 2026

- Project 1: Architected an asynchronous, full-stack LLM/RAG QA audit platform using FastAPI and Next.js, cutting batch audit runtime from 12 hours to 2 hours (83% reduction) for 2,000+ daily cases via rolling queues, connection pooling, and adaptive concurrency control.
- Project 2: Built a distributed, constraint-based heuristic workforce scheduling engine with multi-phase optimisation and automated repair algorithms, dynamically allocating 100+ personnel across multi-tier shifts under complex operational and capacity limits. Allocation fairness increased from 12.5% to 75%.

Republic of Singapore Navy - Naval Officer

Mar 2023 - Jan 2024

- Led a team of 30+ servicemen, ensuring discipline, safety, and mission readiness in high-pressure environments.
- Developed leadership, crisis management, and execution skills transferable to fast-paced technical teams.

Projects

Live Cinema Aggregator - Vue.js, JavaScript, Playwright, Vite, Vercel

May 2026 | cinemagowhere.vercel.app

- Built and deployed a live cinema aggregator consolidating all Shaw and GV showtimes into a searchable web app.
- Developed a live Playwright scraper to extract, clean, and normalise real-time showtime and booking data.
- Implemented movie and cinema views with date filtering, session grouping, and direct booking redirects using Vite.

Scalable Travel Workflow Engine - Vue.js, Firebase, Authentication, Realtime Database

Mar 2026 | zetto.site

- Architected a Firebase backend for complex relational datasets with real-time sync and secure authentication.
- Built a Vue.js interface to visualise nested data structures and streamline multi-step travel workflows.
- Reduced workflow bottlenecks through a low-latency, user-facing data management system.

AMZN Stock Price Forecasting - Python, XGBoost, LSTM, Pandas, NumPy

May 2025 - Jun 2025 | lucasasz.com

- Developed machine learning models to forecast AMZN stock price movement over a 14-day horizon.
- Preprocessed financial time-series data and visualised trends using Matplotlib and Seaborn.
- Compared XGBoost and LSTM performance to demonstrate applied ML techniques in financial analytics.

Credit Card Fraud Detection Model - Python, SQL, Scikit-learn, SHAP, Streamlit

Jun 2025 | fraud-detection-model-lucasang.streamlit.app/

- Built an end-to-end fraud detection pipeline using engineered SQL features and a RandomForest classification model.
- Evaluated model performance using ROC/PR curves and SHAP interpretability; achieved 90%+ ROC-AUC.
- Deployed an interactive Streamlit dashboard for user-facing fraud risk analysis.

Singapore Premier League Performance Analytics - Python, Pandas, NumPy, Matplotlib, Plotly, Notion

Jun 2025 | notion.site

- Analysed Singapore Premier League match data to identify player and team performance trends.
- Built static and interactive visualisations to translate raw data into clear performance insights.
- Authored a structured analysis report covering methodology, findings, and implications for player evaluation.

Skills

Languages: Python, JavaScript, SQL, R, HTML/CSS

Frameworks: FastAPI, Flask, Next.js, Vue.js, React, Scikit-learn, TensorFlow/Keras, XGBoost, Streamlit, TailwindCSS

Data & Visualisation: Pandas, NumPy, Vector Databases, RAG/LLM Pipelines, SHAP, Matplotlib, Plotly, Tableau, Power BI

Databases & Cloud: MySQL, PostgreSQL, AWS, Azure

Tools: Git, Playwright, Vite, Vercel, Render, DBeaver, Jupyter